

WYATT SHORE

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Mechanical engineering student with 16 months of co-op designing, building and testing precision electromechanical systems — from CAD and machining through instrumented validation, embedded firmware and motor control.

EXPERIENCE

3DM DEVICES

Aldergrove, BC

Mechanical Design & Test Engineer — 3 co-op terms

May–Sept 2024 | May–Dec 2025 | May–Sept 2026

- **Owned a dual-purpose pneumatic handling arm from concept to validated prototype**, replacing two mechanisms with one: linear slides, twin 3/4" cylinders, machined aluminum and sheet metal. Now the basis for production units.
- **Designed and built the lab test equipment to prove it out** — an Arduino rig with LiDAR, load cell, pressure transducer and a custom slider-crank position sensor, plus a Python application for live capture and automatic test reports.
- **Ran the validation program** on impact timing, cushioning and repeatability across tire types; reported honestly that my own closed-loop cushion control failed to beat a fixed valve opening, and kept the simpler design.
- **Redesigned a cam-driven scan head trajectory as an optimized polynomial motion profile**, ending follower separation and vibration; adopted into the product and re-engineered for CNC machining from aluminum flat bar.
- **Prototyped a handheld laser scanner end to end**: linear CCD to Arduino, nanosecond timing verified on an oscilloscope, MOSFET level shifting, packaged with battery, display and trigger in a 3D-printed enclosure.
- **Designed a sheet-metal air-purge shield** to keep debris off optical glass in a factory environment, driving it from cardboard airflow mock-ups through outsourced fabrication to installation ahead of a shipping deadline.
- **Held $\pm 5\ \mu\text{m}$ optical stability across a 40 °C range** through thermal expansion analysis on precision mounts, and moved a scanner lid from cast iron to aluminum for heat dissipation.
- **Cut required pressure regulators from 4 to 1** on a pneumatic actuator assembly by resizing cylinders and revising linkages; sourced parts and worked daily with in-house machinists on DFM.

PROJECTS

ROTARY INVERTED PENDULUM

Personal Mechatronics Project

CAD → fabrication → firmware → system ID → optimal control

Jan – Aug 2026

- **Designed, built and programmed a desktop self-balancing pendulum on a \$150 budget** (SolidWorks, 3D-printed structure, slip ring, dual SPI encoders) that swings itself upright from rest and balances indefinitely within $\pm 0.33^\circ$.
- **Wrote field-oriented control for the brushless motor from scratch in C++ on an ESP32**, with no motor library: SPI encoder decoding, startup calibration, 20 kHz three-phase commutation, cascaded position and velocity loops.
- **Derived the equations of motion by Lagrangian mechanics** and measured the real plant on hardware (inertia within 1.3% of theory); designed an LQR controller and state observer to fit the motor's measured 0.087 N·m torque budget.
- **Built a Bluetooth tuning GUI** with live plots and gain sliders to replace firmware reflashes; firmware and GUI published on GitHub, full build log on my portfolio.

SMART BIRDFEEDER — Personal project

Aug – Dec 2025

- Built a solar-powered Raspberry Pi camera system, sealing the housing with O-rings and silicone and fitting a camera inside the seed hopper; wrote a desktop app for image triggers and streaming.

UVIC FORMULA HYBRID — Powertrain team

Jan – April 2025

- Developed an electronic throttle actuator using a stepper motor and rotary encoder; logged motion data through Python into MATLAB and root-caused a steady-state error.

EDUCATION

UNIVERSITY OF VICTORIA — B.Eng. Mechanical Engineering | GPA 8.79 / 9.0

Sept 2024 – Present

UNIVERSITY OF THE FRASER VALLEY — Engineering Transfer | GPA 4.1 / 4.33

Sept 2022 – April 2024

- **Awards:** UVic President's Scholarship, Clara Evelyn Wilson Scholarship, UVic Transfer Entrance Scholarship, UFV Dean's List (3×).

TECHNICAL SKILLS

- **Design:** SolidWorks (parts, assemblies, sheet metal, drawings), AutoCAD, DFM, tolerance stack-up, thermal expansion analysis, sealing and enclosure design, cam and motion profiling.
- **Analysis:** Statics, dynamics, kinematics, thermodynamics, materials selection; Lagrangian modelling, LQR and state observers, PID, field-oriented motor control, MATLAB, Simulink.
- **Build & Test:** Manual mill and lathe, 3D printing, press-fitting, soldering and wiring harnesses; design of instrumented test rigs, oscilloscope, CMM, micron-level laser calibration, factory acceptance testing.
- **Software:** Python (PySide6, Streamlit, NumPy), C++ / Arduino / ESP32, MATLAB, Galil motion control, Git.